

Improvement in Q-factor and Resonance Frequency Depending on the Particle Sizes of Fe-Si-Cr Alloy

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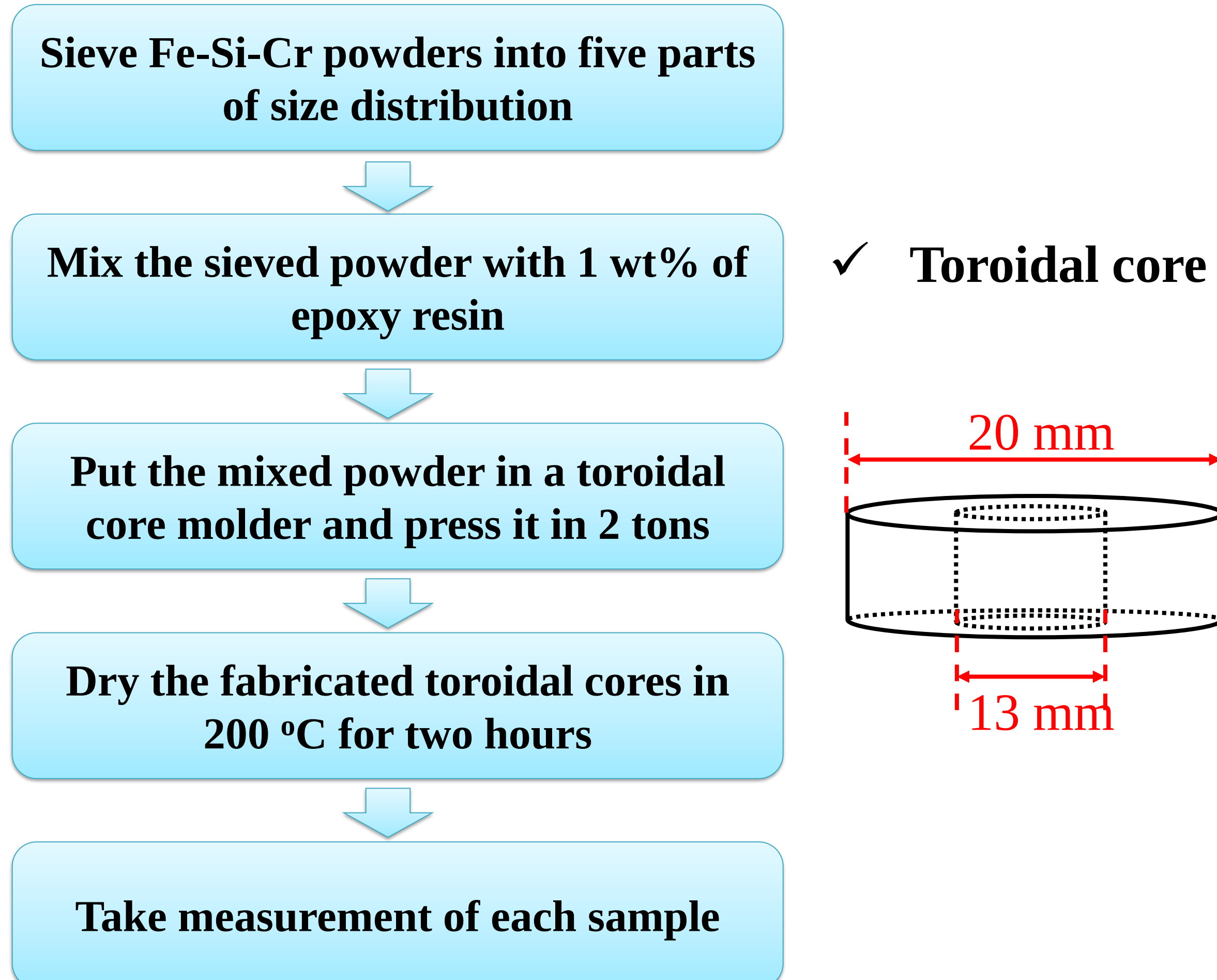
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Introduction

❖ Motivation

- ✓ Soft magnetic composites (SMCs) with high saturation magnetization (M_s), low coercivity (H_c), and low core-loss are essential for power-inductor to achieve high energy efficiency at high frequencies.
- ✓ One of the evaluation methods of power-inductor is measurement of Q-factor, which is related to magnetic permeability and resistivity.
- ✓ The particle size is related to the powder's resistivity, and affects on the density of fabricated core, which is related to permeability.
- ✓ We made toroidal core type samples depending on particle sizes, and measured permeability and Q-factor to check if the core is suitable for high-frequency application.

❖ Preparation of FSC powders and Toroidal core



❖ Permeability & Q-factor

✓ Permeability

$$\mu = \frac{B}{H} = \frac{B_0 e^{j(\omega t - \delta)}}{H_0 e^{j\omega t}} = \frac{B_0}{H_0} \cos \delta - j \frac{B_0}{H_0} \sin \delta = \mu' - j\mu''$$

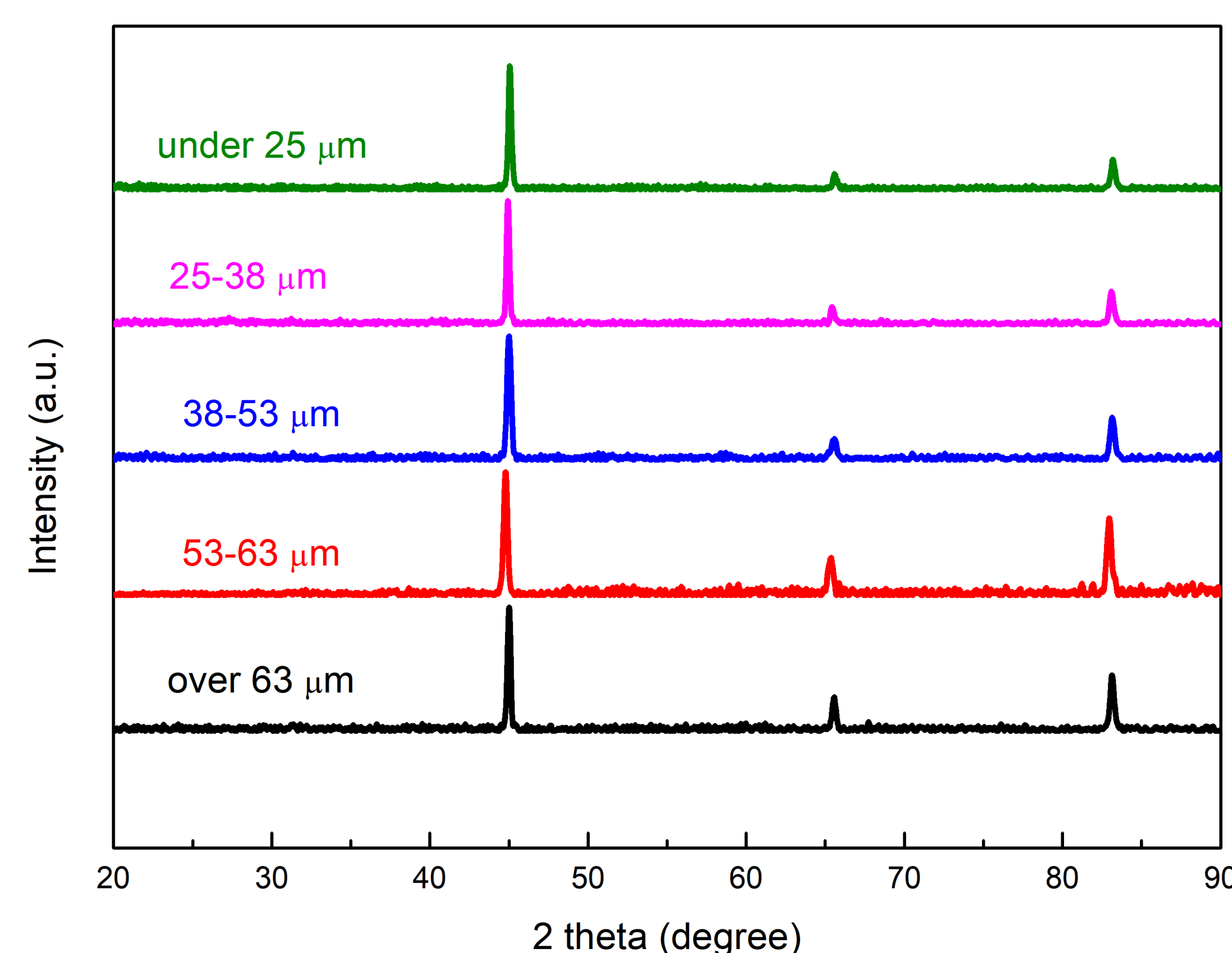
✓ Q-factor

$$Q = \frac{2\pi f L}{R_s} = \frac{\mu'}{\mu''} = \frac{1}{\tan \delta}$$

- ✓ In order to gain high Q-factor in a particular frequency, high permeability is needed in that frequency.

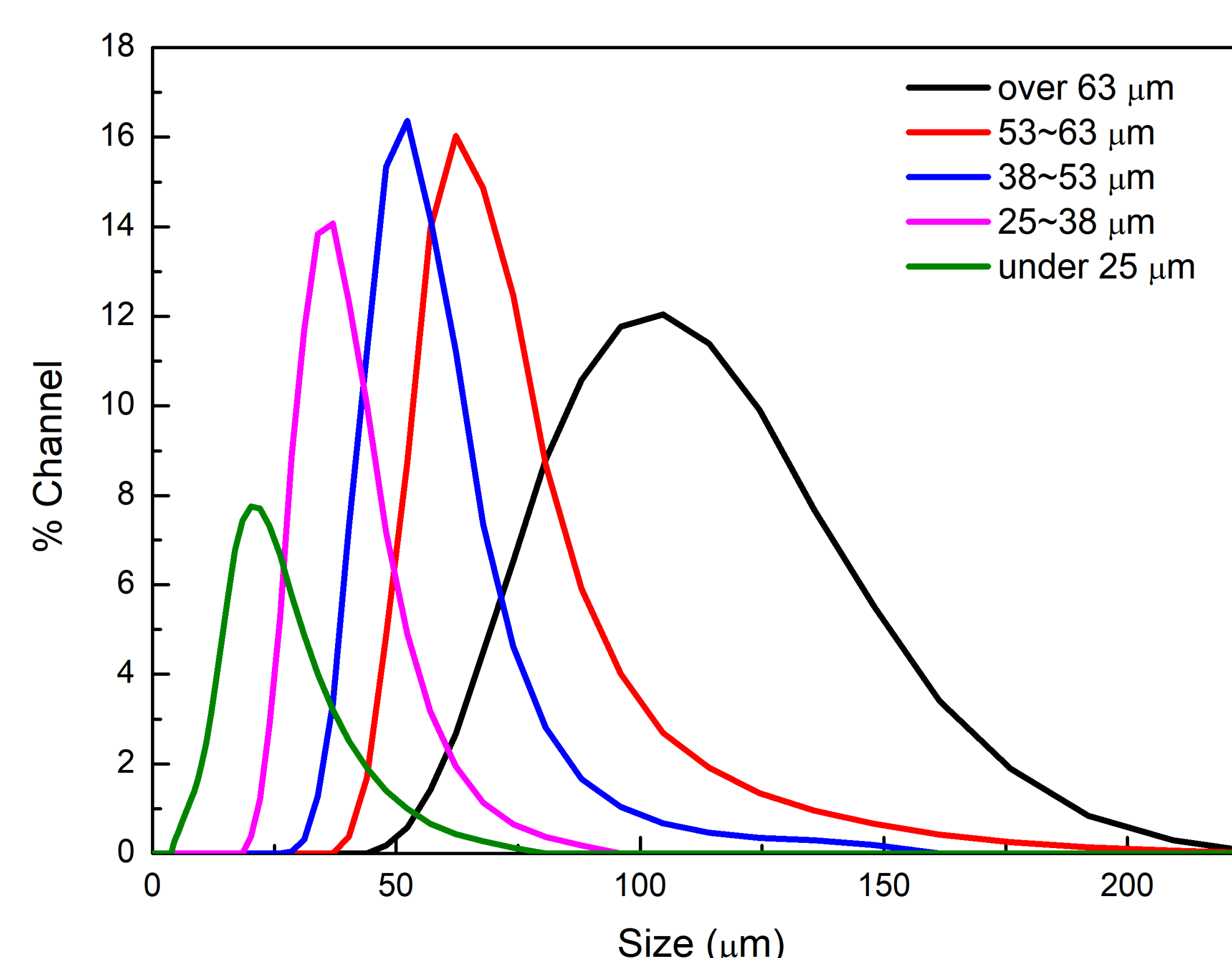
Experiments & Results

❖ X-ray Diffraction (XRD)



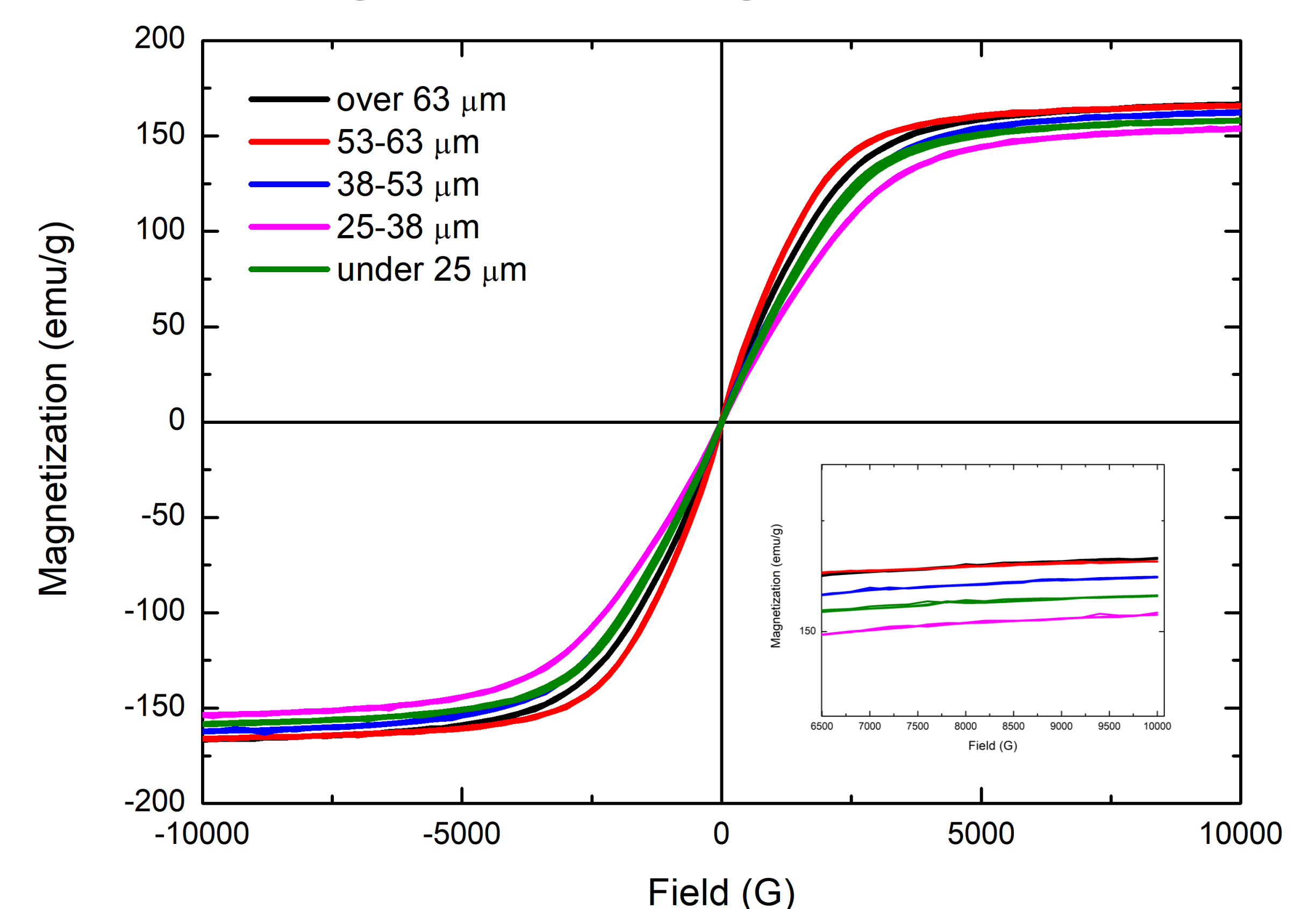
- ✓ All powders show same structure as simple cubic α -Fe structure.
- ✓ We can know that the main element of the alloy is Fe.

❖ Particle Size Distribution (PSD)



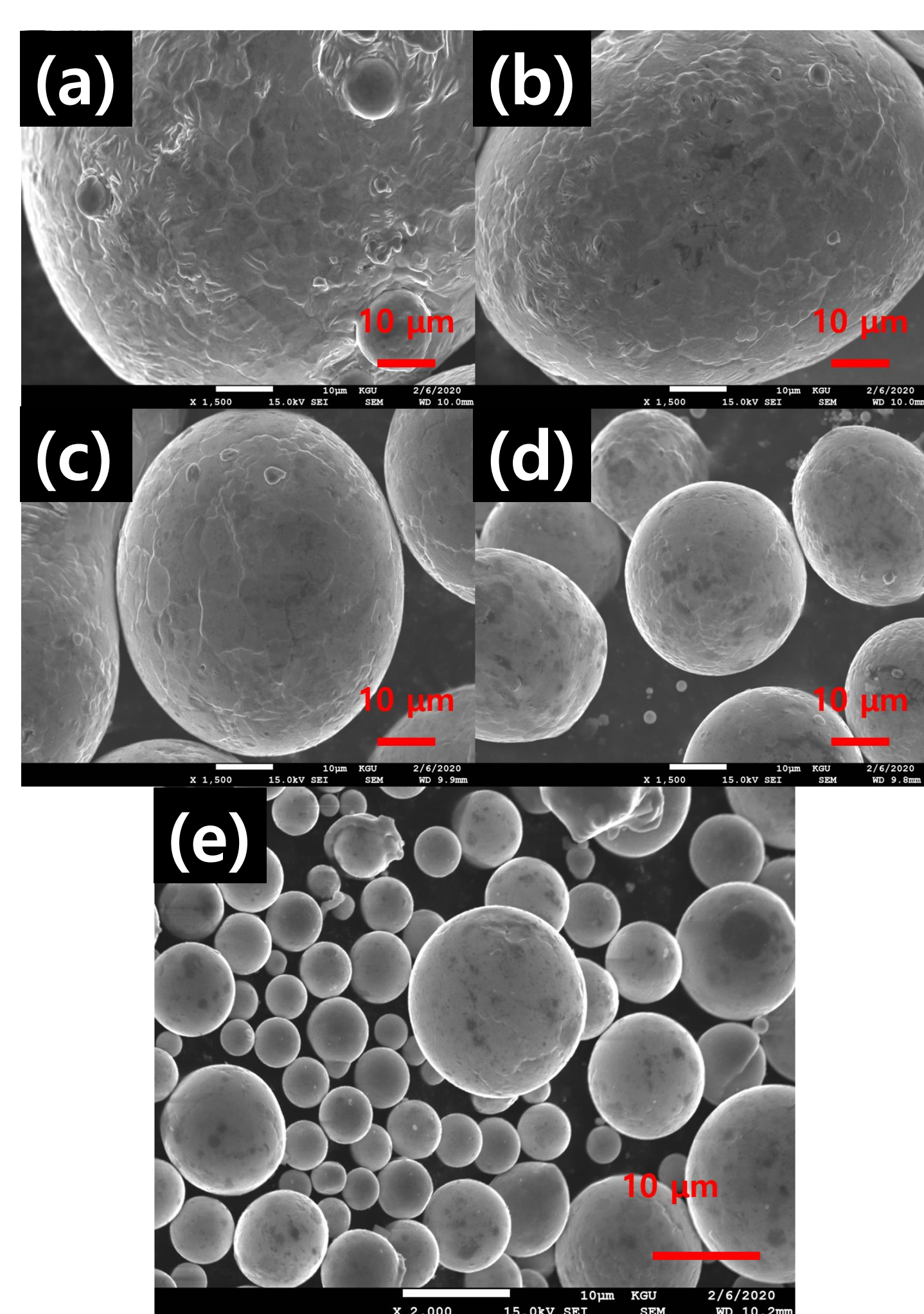
- ✓ We can see that the distribution in most samples is uniform
- ✓ However, over 63 μm shows quite a wide distribution

❖ Vibrating Sample Magnetometer (VSM)



- ✓ All the samples with various particle sizes show slight difference in magnetization saturation and coercivity.
- ✓ However, their differences are not that significant, and show a soft magnetic property.

❖ SEM & EDS

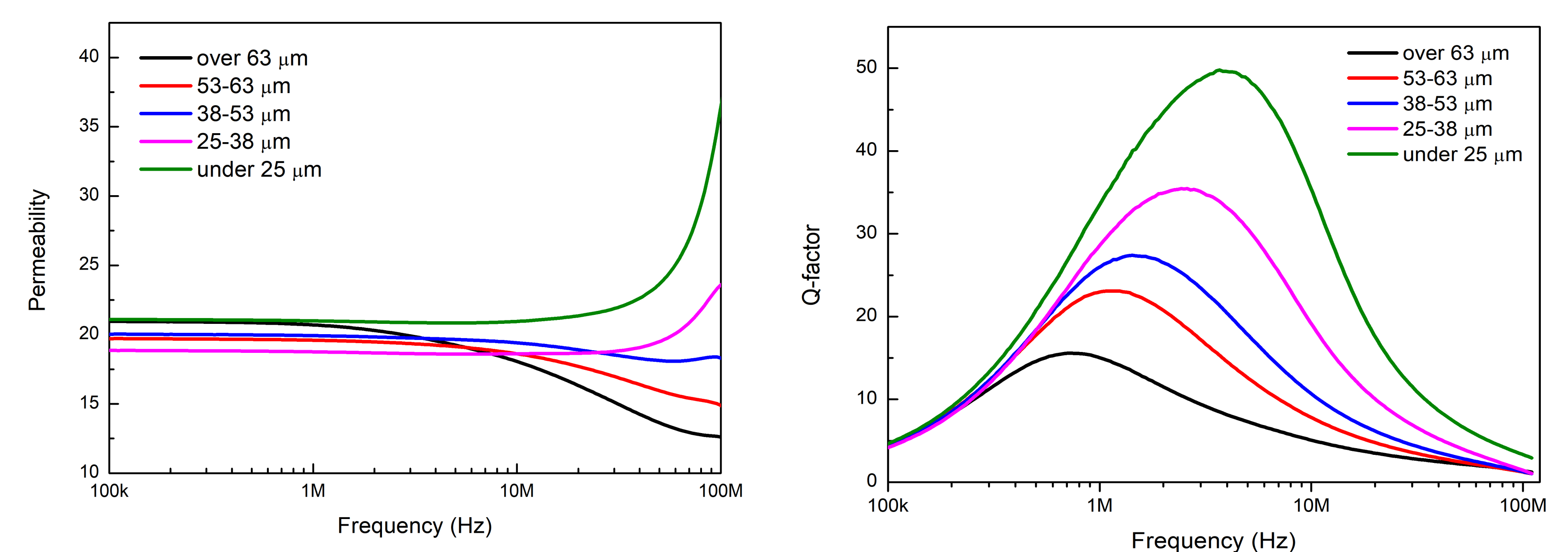


	< 25 μm	25-38 μm	38-53 μm	53-63 μm	> 63 μm
Fe	86.45 wt%	86.22 wt%	87.79 wt %	87.31 wt %	87.17 wt%
Si	11.39 wt%	11.33 wt%	10.51 wt%	10.08 wt%	10.39 wt%
Cr	2.17 wt%	2.45 wt%	1.70 wt%	2.61 wt%	2.45 wt%

- ✓ SEM image shows that Gas-atomized powders exhibit spherical morphology.
- ✓ The EDS table insists that the average amount of each element included in the alloy is 87 wt%, 11 wt%, and 2 wt% of Fe, Si, and Cr elements, respectively.

Morphology of Fe-Si-Cr powders (a) over 63 μm (b) 53-63 μm (c) 38-53 μm (d) 25-38 μm (e) under 25 μm

❖ Permeability & Q-factor



- ✓ The powder with under 25 μm shows the highest permeability in 1 MHz, and 3 MHz, which are the most used bands for electronic devices.
- ✓ Powders which their permeability rapidly decreases in high-frequency are not suitable for high-performance electronic devices.
- ✓ The Q-factor tendency shows that it increases as powder size decreases.
- ✓ Also, the resonance frequency increases as powder size decreases, with 49.8 of Q-factor, and 3.68 MHz of resonance frequency at under 25 μm sample.

Conclusions

- ✓ In this study, the permeability, and Q-factor dependency of various cores with different particle sizes were investigated.
- ✓ Changes in permeability and Q-factor according to the particle sizes of the samples were found.
- ✓ The sample with smallest particle size showed the highest permeability and Q-factor.
- ✓ Especially, Q-factor and resonance frequency show a uniform tendency that as the particle size decreases, they increases gradually, and the highest value is achieved at under 25 μm with 49.8 of Q-factor, and 3.68 MHz of resonance frequency.